Heng Yang

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Research Interests

Robot Dynamics, Control Systems, Motion Planning, Mechatronic Systems, Dynamic Modeling

Education

University of Pennsylvania Philadelphia, PA
M.S. in Mechanical Engineering and Applied Mechanics *Aug. 2025 – May 2027*

- GPA: 4.00/4.00
- Relevant Coursework: Design of Mechatronic Systems (A), Introduction to Robotics (A), Performance, Stability and Control of UAVs (A), Advanced Dynamics, Control Optimization for Robotics

The University of Arizona Tucson, AZ
B.S. in Mechanical Engineering, Summa Cum Laude *Sep. 2021 – Jul. 2025*

- GPA: 3.95/4.00; Outstanding Senior Nominee
- Relevant Coursework: Control Theory, Mechanical Vibrations, Dynamics and Control Lab, Numerical Methods, Finite Element Methods, Fluid Mechanics, Thermodynamics, Heat Transfer

Hebei University of Technology Tianjin, China
B.Eng. in Mechanical Engineering *Sep. 2021 – Jul. 2025*

- GPA: 3.93/4.00; Ranked 3rd out of 70 students

Research Experience

ModLab, University of Pennsylvania Philadelphia, PA
Research Assistant / Graduate Researcher *May 2026 – Present*

- Designed and fabricated an experimental platform for characterizing twisted-string actuators under varying payload and mechanical constraint conditions to optimize the life cycle of string and actuator.
- Developed custom CAD components including weight carriages, encoder mounts, linear-bearing supports, actuator fixtures, and modular sensor holders.
- Integrated AS5600 magnetic encoders, INA260 current sensors, infrared limit sensors, and ESP32-S3 microcontrollers into a synchronized sensing and control system.
- Developed Arduino-based data acquisition software for magnetic field strength, logging actuator angle, current, voltage, and cycle count during repeated actuation experiments.
- Investigated the effects of magnetic alignment, air-gap variation, and sensor placement on encoder reliability and measurement repeatability.

Helicopter Trim Solver and Sensitivity Analysis Apr. 2026 – Jun. 2026

- Developed a nonlinear helicopter trim solver in MATLAB for steady forward-flight analysis.
- Implemented iterative inflow calculation, rotor flapping dynamics, aerodynamic coefficient estimation, and six-equation force/moment equilibrium.
- Performed sensitivity studies on center-of-gravity placement, horizontal tail configuration, tail rotor position, and flapping frequency.
- Evaluated the effects of design parameters on control input, aircraft attitude, rotor flapping response, and power requirement.

Rocket Engine CFD Simulation and Analysis Apr. 2026 – Jun. 2026

- Designed methane–oxygen rocket engine geometry in ANSYS DesignModeler, including the combustion chamber and converging–diverging nozzle.
- Simulated internal reacting flow in ANSYS Fluent to analyze pressure, temperature, velocity distribution, and nozzle acceleration.
- Applied turbulence modeling, energy equations, mesh independence studies, and convergence analysis to improve solution stability and result reliability.

Selected Robotics and Control Projects

Franka Emika Panda Robot Manipulation Sep. 2025 – Dec. 2025

- Developed a complete Python software stack for a 7-DOF Franka Panda manipulator, including forward kinematics, inverse kinematics, and trajectory generation.
- Implemented Jacobian pseudoinverse- and Jacobian transpose-based inverse kinematics with null-space optimization for redundant manipulator control.
- Developed Rapidly-exploring Random Tree and Artificial Potential Field motion planners with collision checking for autonomous manipulation tasks.
- Generated Cartesian line, circular, and elliptical trajectories using differential inverse kinematics and evaluated tracking behavior in simulation.
- Integrated vision-based object localization with ROS/RViz to autonomously pick, transport, and stack square blocks.

Autonomous Differential-Drive Mobile Robot Oct. 2025 – Dec. 2025

- Designed and assembled a differential-drive mobile robot using dual ESP32 microcontrollers, encoder motors, ToF sensors, ultrasonic sensing, and Vive localization.
- Implemented closed-loop PID velocity control using encoder feedback to regulate motor speed under varying load and terrain conditions.
- Integrated ToF and ultrasonic sensors for obstacle avoidance and wall-following navigation on a 12 ft $\times$ 5 ft track with a ramp.
- Developed a Wi-Fi-based remote control interface with real-time telemetry visualization and online PID parameter tuning.
- Reduced lap completion time by 18% through controller tuning, sensing improvement, and mechanical adjustment.

Bachelor Thesis: Design and Simulation of an AGV System with Manipulator Jan. 2025 – Jun. 2025

- Designed an Automated Guided Vehicle integrated with an RPRR robotic manipulator for material handling applications.
- Performed torque analysis, structural design, stress analysis, and strength verification for key mechanical components.
- Developed an A*-based global path planning framework with weighted heuristics, improving computational efficiency by 15% while maintaining path optimality.
- Evaluated manipulator workspace, payload capability, and AGV navigation performance through simulation and analytical modeling.

National University Physics Experiment Competition Jul. 2022 – Nov. 2022

- Constructed an experimental optical system to analyze the relationship between temperature and solution concentration.
- Processed and visualized experimental data in MATLAB and summarized experimental findings in a 40-page technical report.
- Awarded National Second Prize in the 8th National College Students Physics Experiment Innovation Competition.

Industry Experience

Xiamen ABB Low Voltage Electrical Equipment Co., Ltd. Xiamen, China
Research and Design Intern *Jul. 2024 – Sep. 2024*

- Analyzed experimental current data for low-voltage electrical products using numerical methods and data-processing workflows.
- Identified indexing errors in large experimental datasets and improved data reliability using KNIME.
- Assisted with prototype assembly, supplier coordination, and experimental preparation for product validation.
- Developed a 20-page user manual covering product parameterization, usage, and precautions.

Langfang Jingdiao CNC Machine Tool Manufacturing Co., Ltd. Langfang, China
Production Intern *Jan. 2024 – Feb. 2024*

- Modeled mechanical parts and generated machining paths using CAD/CAM software for precision manufacturing.
- Designed a fingertip gyro model, simulated tool paths, and wrote machining code for precision engraving.

Patents and Software Copyright

- Lixin Sun, Guanjie Mao, Yilan Zhang, Muyuan Yang, **Heng Yang**, et al. *An Umbrella Drying Device*, Chinese Utility Model Patent, ZL202322873089.X, Jun. 2024.

- Wei Liu, Guanjie Mao, Jingsong Wei, Yilan Zhang, **Heng Yang**, et al. *A Somatosensory Controller Holder*, Chinese Utility Model Patent, ZL202323057571.2, Jun. 2024.
- Wei Liu, Guanjie Mao, Muyuan Yang, **Heng Yang**, et al. *A Support for Leap Motion*, Chinese Design Patent, ZL202330760025.1, Jul. 2024.
- *Mechanical Simulation Virtual Training Service System V1.0*, Computer Software Copyright, 2023R1704772, Dec. 2023.

Honors and Awards

- Outstanding Senior Nominee, The University of Arizona, Spring 2025.
- Outstanding Bachelor Thesis, Hebei University of Technology, Spring 2025.
- Dean's List with Distinction, The University of Arizona, Spring 2024 and Fall 2024.
- National Second Prize, 8th National College Students Physics Experiment Innovation Competition.
- Provincial Third Prize, 9th Hebei Province "Internet+" College Student Innovation and Entrepreneurship Competition.

Technical Skills

- **Programming:** Python, MATLAB, C, Arduino, C#, LaTeX
- **Robotics and Control:** PID Control, Motion Planning, Forward/Inverse Kinematics, Trajectory Generation, Sensor Integration, ESP32, RViz
- **Simulation and Modeling:** ANSYS Fluent, ANSYS Workbench, Simulink, Finite Element Analysis, CFD
- **CAD/CAM:** SolidWorks, Onshape, Creo, AutoCAD, Draftsight
- **Tools:** Git, Overleaf, Microsoft Office, KNIME